



File Operations in Python & Word Count Program

Name:- Rutuja Nitin Patil

PRN:-202501040178

DIV:-CS-2

Batch:-C-24

Introduction

- File handling is used to store and manage data permanently.
- Python provides built-in functions to work with files.
- It helps in reading ,writing and updating data efficiently.



File Operations

- 01 Open a file using `open()`
- 02 Read data using `read()`
- 03 Write data using `write()`
- 04 Append data using `append()`
- 05 Close file using `close()`



File Modes

- `r` → Read mode
- `w` → Write mode
- `a` → Append mode
- `r+` → Read and Write mode

Problem Statement

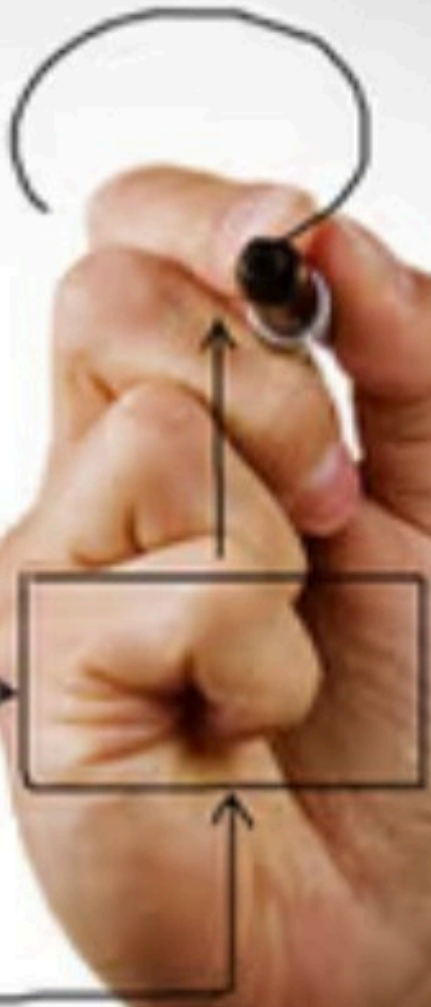


- 01 Need to count total number of words in a text file
- 02 File may contain large amount of text
- 03 Manual counting is time-consuming and error-prone
- 04 Requires an automated solution using Python

Problem Analysis

- 01 Input: Text file
- 02 Process: Read file content
- 03 Output: Total number of words
- 04 Challenge: Handle large text efficiently





- 05

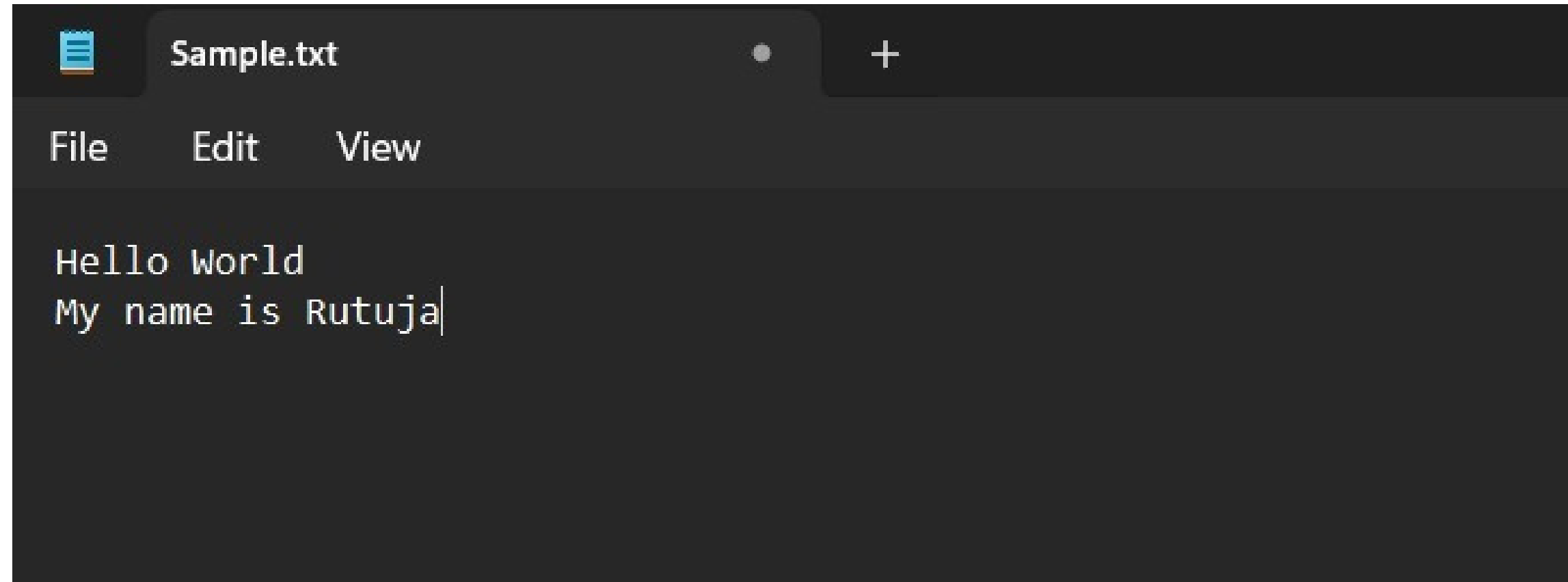
Python Program

```
with open("sample.txt", "r") as file:  
    data = file.read()
```

```
    words = data.split()  
    word_count = len(words)  
    print("Total words:", word_count)
```

```
with open("sample.txt", "r") as file:  
    data = file.read()  
  
words = data.split()  
word_count = len(words)  
  
print("Total words:", word_count)
```

Output



A screenshot of a text editor window. The title bar shows a file icon, the name 'Sample.txt', and a '+' button. The menu bar contains 'File', 'Edit', and 'View'. The text area contains two lines: 'Hello World' and 'My name is Rutuja' with a cursor at the end of the second line.

```
Sample.txt
```

```
File Edit View
```

```
Hello World  
My name is Rutuja|
```

Output

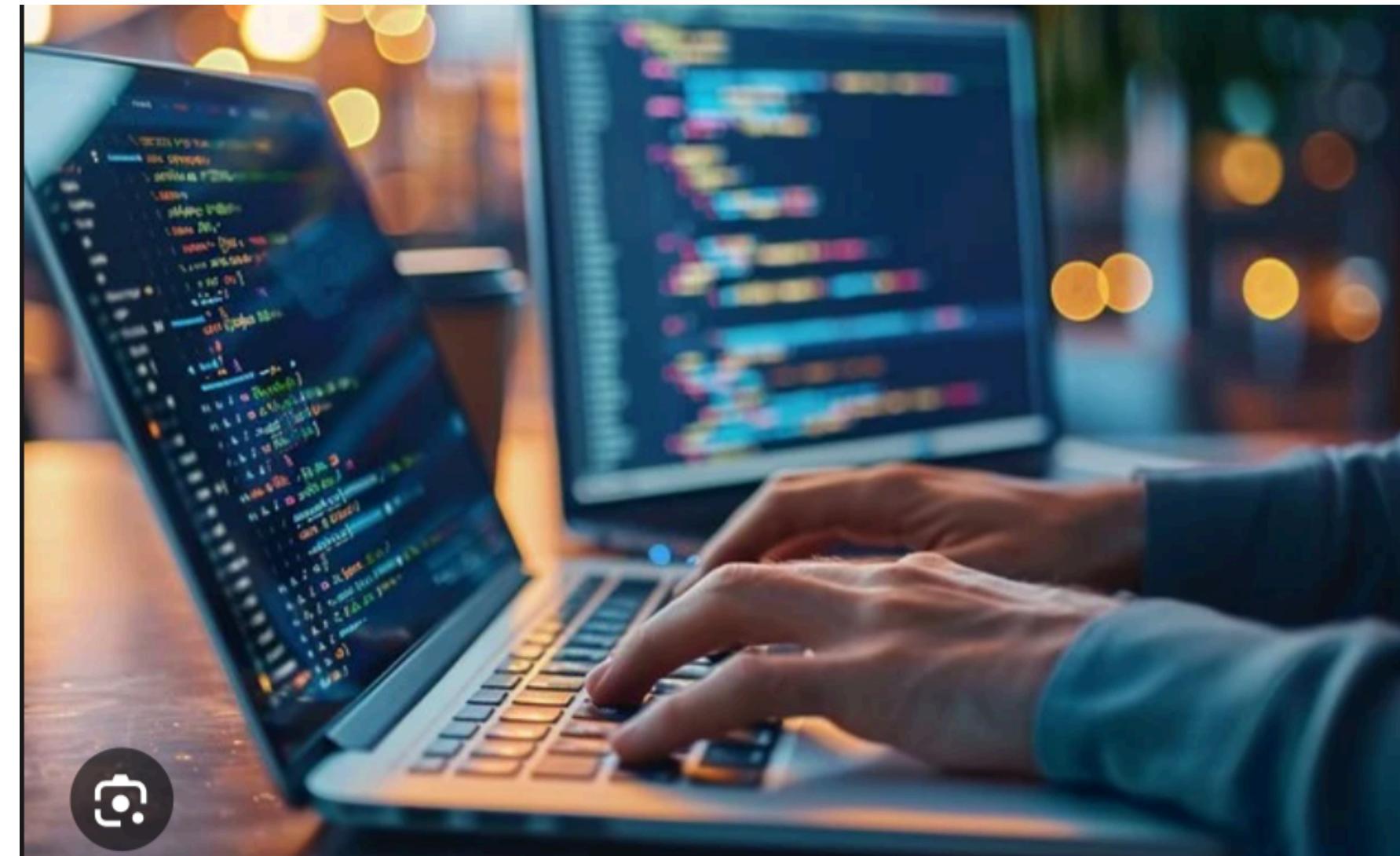
Clear

Total words: 6

=== Code Execution Successful ===

Complex Problem Interpretation

- Handling large files efficiently
- Data may include spaces, symbols, multiple lines
- Need accurate and fast processing
- Solution can be extended for advanced text analysis



Application

- Textfile Analysis
- Content processing
- Report generation
- Data analytics



Conclusion

- File operations are essential in Python
- Word counting is a useful real-world application
- Automation improves efficiency and accuracy

